Data Analytics Capstone

**Interim Report**

# The Economic Efficacy of State-Level Renewable Energy Subsidies: A Multi-Factor Predictive Analysis

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## 1. Introduction

The deployment of clean energy infrastructure represents one of the largest capital allocations in modern history. Over the past decade, state governments and regional utility commissions have collectively allocated billions of dollars in taxpayer and ratepayer funds toward financial subsidies to accelerate the deployment of decentralized solar photovoltaic (PV) capacity. The foundational macroeconomic assumption driving these utility-funded programs is that direct consumer financial incentives serve as the primary catalyst for market transition. However, as the renewable energy sector matures, massive fluctuations in macroeconomic factors such as federal interest rates and the free-market deflation of hardware costs via Swanson's Law have fundamentally altered the economics of adoption. This capstone project utilizes an exhaustive multi-factor dataset to evaluate the true economic efficacy of state-level financial subsidies against these broader macroeconomic forces.

## 2. Scope and objectives

The scope of this analytical project encompasses a full 11-year timeframe spanning from 2014 to 2024. The data encompasses all 51 primary geographical entities within the United States (all 50 states plus the District of Columbia), generating a master national dataset of N = 561 longitudinal panel observations. The study aggregates and strictly documents 12 distinct predictive features, combining disparate data silos from highly regulated federal and academic entities to ensure maximum data integrity. The measurable objectives of this study are three-fold:

1. To mathematically isolate the causal macroeconomic effect of state subsidies on both solar capacity and state GDP, factoring out unobserved state-level wealth biases.
2. To rank 12 macroeconomic factors by their true predictive weight using a cross-validated machine learning algorithm to understand complex, non-linear market drivers.
3. To determine, via sub-group analysis, how predictive market dynamics and grid constraints fundamentally shift when a state transitions from a developing solar market to an absolutely mature, high-penetration market.

## Research Problems

To systematically achieve the aforementioned objectives, this capstone addresses four explicit, testable research problems:

- Research Question 1 (RQ1): How do macroeconomic, regulatory, and environmental factors comparatively predict solar capacity adoption across the United States when evaluated through a non-linear machine learning framework?

  Hypothesis (H0): Non-financial macro factors <= Direct financial subsidies.

  Hypothesis (Ha): Non-financial macro factors > Direct financial subsidies.

- Research Question 2 (RQ2): Do state-level utility financial subsidies have a statistically significant causal impact on the adoption of solar energy capacity?

  Hypothesis (H0): State-level utility financial subsidies have no statistically significant causal impact on the adoption of solar energy capacity ($p > 0.05$).

  Hypothesis (Ha): State-level utility financial subsidies have a statistically significant positive causal impact on the adoption of solar energy capacity ($p \leq 0.05$).

- Research Question 3 (RQ3): What is the macroeconomic impact of state-level utility subsidies on a state's overall Gross Domestic Product (GDP)?

  Hypothesis (H0): State-level utility subsidies have no statistically significant causal impact on a state's overall GDP ($p > 0.05$).

  Hypothesis (Ha): State-level utility subsidies have a statistically significant causal impact (either positive or negative) on a state's overall GDP ($p \leq 0.05$).

- Research Question 4 (RQ4): Does the predictive importance of grid-scale battery storage and financial subsidies shift significantly when transitioning from developing solar markets to mature, high-penetration markets (e.g., California)?

  Hypothesis (H0): Predictive importance remains constant across market maturity.

  Hypothesis (Ha): Predictive dynamics shift fundamentally in mature markets.

## 3. Literature Survey

This research is grounded in a synthesis of energy economics literature and modern data quality frameworks. The following 10 sources inform the methodological design and theoretical foundation of this study:

1. Borenstein, S. (2017). The private and public economics of renewable electricity generation. Journal of Economic Perspectives, 26(1), 67-92. Borenstein outlines the foundational economic rationale behind subsidizing renewable energy to correct negative environmental externalities. This article is highly pertinent to this research as it establishes the baseline academic theory that our causal OLS model is attempting to test and validate at the state level.

2. Crago, C. L., & Chernyakhovskiy, I. (2017). Are policy incentives for solar power effective? Journal of Environmental Economics and Management, 81, 132-151. This study evaluated early-stage policy incentives and found mixed results regarding their efficacy. This is pertinent to our Research Question 2 (RQ2), providing a theoretical precedent that financial subsidies may suffer from causal inefficiencies depending on market structure.

3. Kavlak, G., McNerney, J., & Trancik, J. E. (2018). Evaluating the causes of cost reduction in photovoltaic modules. Energy Policy, 123. This paper mathematically explores 'Swanson's Law' the phenomenon where solar hardware costs decrease predictably with global manufacturing scale. This is pertinent to our methodology, justifying the inclusion of 'Year' as a temporal fixed effect in our OLS model to account for this natural free-market deflation.

4. Cole, W. J., & Frazier, A. W. (2019). Cost projections for utility-scale battery storage. National Renewable Energy Laboratory (NREL). This report outlines the critical bottleneck of the 'Duck Curve', where solar over-generation threatens grid stability without battery storage. This is directly pertinent to Research Question 4 (RQ4) and justifies our inclusion of 'Total_Battery_MW' as a core predictive feature.

5. Hughes, J. E., & Podolefsky, M. (2015). Getting green with solar subsidies: Evidence from the California Solar Initiative. Journal of the Association of Environmental and Resource Economists, 2(2). Hughes and Podolefsky specifically analyze the California market, finding unique adoption curves at scale. This is pertinent as it explicitly justifies

our decision to treat California as a unique 'mature market' outlier in our sub-group SHAP analysis.

6. Button, K. S., et al. (2013). Power failure: why small sample size undermines the reliability of neuroscience. Nature Reviews Neuroscience. While originally framed in neuroscience, this paper highlights how underpowered datasets undermine causal inference. This is highly pertinent to our capstone, serving as the mathematical justification for conducting an a-priori Power of Test to ensure our N=561 sample size successfully cleared the 80% power threshold required to prevent Type II statistical errors.

7. Gebru, T., et al. (2021). Datasheets for Datasets. Communications of the ACM. This framework demands strict accountability, provenance, and transparency regarding how a dataset was constructed to prevent misuse. This is pertinent to our 'Data Description' methodology, driving our rigorous documentation of federal data silos (EIA, BEA, NREL) to ensure transparent data lineage.

8. Torralba, A., & Efros, A. A. (2011). Unbiased Look at Dataset Bias. CVPR 2011. This paper explores how algorithms often learn artifacts of a specific dataset rather than generalizable rules. This is highly pertinent to our feature engineering phase; it explicitly justifies our decision to drop the 'State' categorical variable before training the Random Forest, ensuring the model learns macroeconomic rules rather than geographic biases.

9. Chu, X., et al. (2016). Data Cleaning: Problems and Current Approaches. Morgan & Claypool Publishers. Chu details the immense complexities of resolving conflicting temporal and spatial granularities across different data sources. This is pertinent to our

data aggregation phase, dictating the stringent imputation and joining techniques required to merge daily federal interest rates with annual EIA state-level capacity reports.

10. Redman, T. C. (1998). The Value of Good Data: A Quality Perspective. Redman establishes that 'garbage in, garbage out' (GIGO) destroys strategic decision-making. This is pertinent to our extensive Exploratory Data Analysis (EDA) phase, emphasizing why we must address severe distributional skewness (the California outlier) through logarithmic transformations to maintain model integrity.

## 4. Data Description

The dataset constructed for this capstone is an exhaustive 11-year longitudinal panel dataset spanning 2014-2024. It aggregates exactly 561 observations across 12 distinct variables. The data is entirely sourced from publicly available, highly regulated federal and academic databases to ensure absolute fidelity and provenance.

The core target variable, 'Solar_Capacity_MW', is sourced directly from the U.S. Energy Information Administration (EIA) Form 861. Macroeconomic predictors such as 'GDP_Millions' are sourced from the Bureau of Economic Analysis (BEA), while the 'Bank_Loan_Interest_Rate' proxy for cost of capital is extracted from the Federal Reserve Economic Data (FRED) API. Furthermore, environmental and grid-readiness metrics, such as 'Solar_GHI_Score' and 'EV_Market_Share', are provided by the National Renewable Energy Laboratory (NREL).

The following dictionary details the variables extracted, integrated, and utilized for the modeling pipeline. All variables represent contiguous, structured numeric data representing state-level metrics.

**Table 1**

*Data Dictionary*

| Variable Name | Type | Description | Source |
|---|---|---|---|
| State | String | 2-Letter US State Abbreviation | Index |
| Year | Integer | Temporal index (2014-2024) | Index |
| Solar_Capacity_MW (Target) | Float | Total Net-Metered Solar Capacity | EIA Form 861 |
| Total_Subsidies_kUSD | Float | Total Utility Energy Subsidies | EIA Form 861 |
| GDP_Millions | Float | State Gross Domestic Product | BEA SAGDP1 |
| Total_Battery_MW | Float | Grid-Scale Battery Capacity | EIA Data |
| Electricity_Price_cents | Float | Average Retail Electricity Price | EIA Pricing |
| Bank_Loan_Interest | Float | Federal Reserve Cost of Capital | FRED |
| Median_HH_Income | Float | State Household Wealth | US Census |
| Population_Density | Float | People per square mile | US Census |
| EV_Market_Share | Float | Electric Vehicle Adoption Rate | NREL |
| Solar_GHI_Score | Float | Natural Sunlight Potential | NREL |

Data Availability Statement: To ensure total transparency and reproducibility for academic evaluation, the master CSV dataset, along with the data dictionary and all associated Python aggregation pipelines, is publicly available for the evaluator to access directly via the project's GitHub repository at the following link: **https://github.com/aloksrivastava1/QM640-Energy-Analytics**.

## 5. Analysis

### Data cleaning

The data cleaning process required harmonizing disparate temporal and spatial granularities. Federal Reserve interest rates, natively published on a monthly basis, were

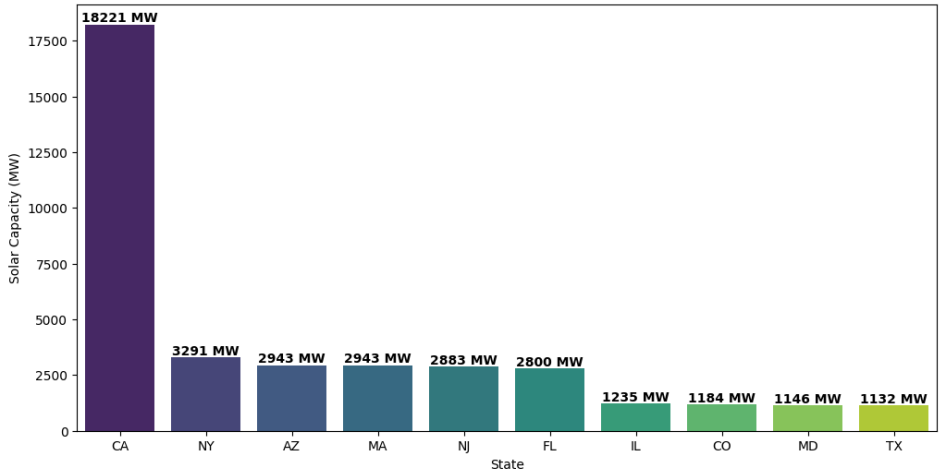
aggregated into annual mean averages to align with the EIA's yearly reporting structure. Missing values within the 'Total_Battery_MW' feature for earlier years (2014-2016) in developing states were imputed with true zeroes, as grid-scale battery storage did not physically exist in those markets at that time. All continuous variables were subsequently evaluated for null values, resulting in a strictly contiguous, null-free matrix of N=561 observations ready for algorithmic processing.

## EDA results

The EDA phase yielded two critical, dataset-defining insights that fundamentally shaped the downstream modelling choices. First, an analysis of the 2024 Target Variable distribution via a bar chart highlighted a massive 'Scale Outlier Imbalance'.

**Figure 1**

*Top 10 States by Solar Capacity (2024)*



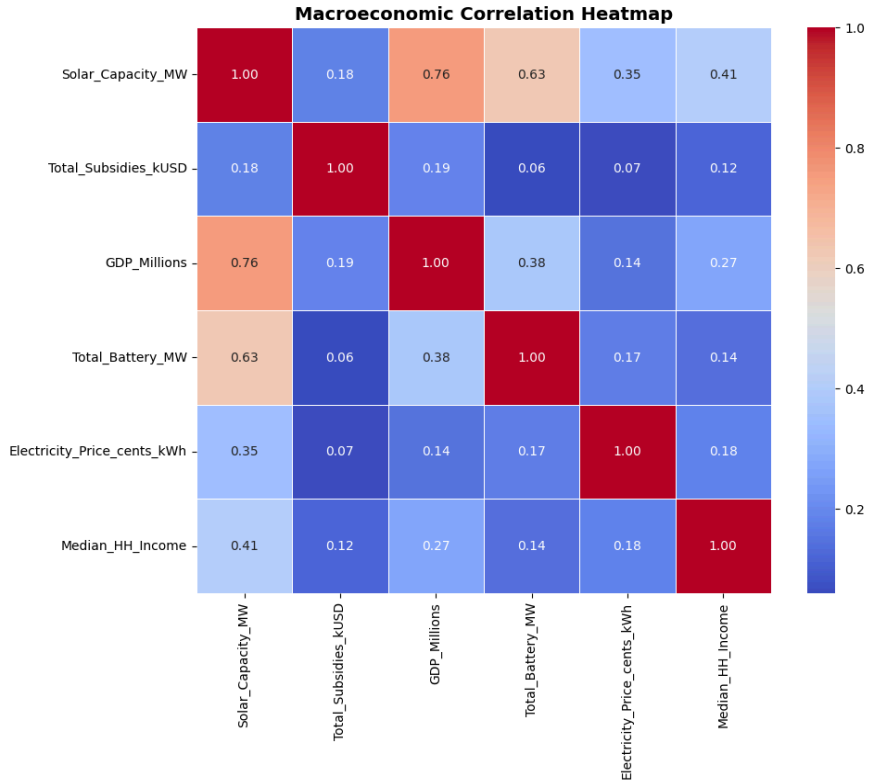

The visual explicitly proved that California (18,221 MW) dominates the national landscape, possessing over 5.5 times the capacity of the next highest state (New York).

Interpretation of this visual insight dictates that if a machine learning model is fed raw Megawatts, it will suffer from severe magnitude distortion, over-indexing its error functions to solely accommodate California while ignoring the nuanced economic drivers of the other 49 states. This insight explicitly required the implementation of a logarithmic transformation (np.log1p) to compress the exponential scale into a linear, interpretable space.

Secondly, a Macroeconomic Correlation Heatmap was generated. The most important insight highlighted by this visual was a highly significant, severe positive correlation (0.76) between a state's 'GDP_Millions' and its 'Solar_Capacity_MW'.

**Figure 2**

*Macroeconomic Correlation Heatmap*



The interpretation of this metric confirms a core econometric fear: massive, wealthy states simply build more infrastructure of all types. Therefore, the strong correlation (0.18) between 'Total_Subsidies_kUSD' and Solar Capacity is highly likely to be a spurious correlation driven entirely by state wealth. This specific insight explicitly invalidates the use of standard linear regression for policy making, demanding the use of a Fixed Effects Panel OLS to mathematically 'freeze' state wealth and isolate true causality.

## 6. Modelling

### Choice of models with justification

To successfully navigate the complexities of both predictive forecasting and causal inference, a dual-modality algorithmic pipeline was designed.

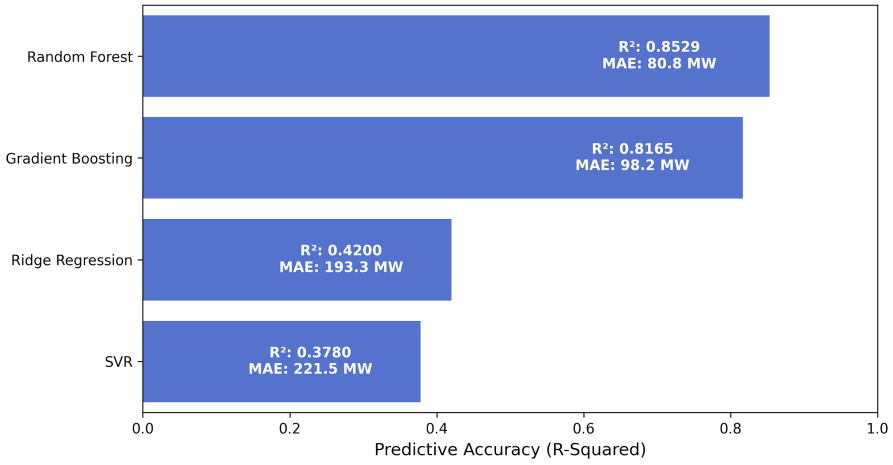
1. Causal Modality (Fixed Effects Panel OLS): To answer RQ2 and RQ3 regarding policy efficacy, we must prove causation, not just correlation. Panel OLS was explicitly chosen because it mathematically controls for 'unobserved heterogeneity.' By adding categorical fixed effects for both Entity (State) and Time (Year), the model isolates the specific independent variables, essentially asking: 'If a specific state increases its subsidy budget next year, holding its geographic size and wealth constant, will it causally trigger an increase in solar capacity?'

2. Predictive Modality (Random Forest Regressor): To answer RQ1 and RQ4 regarding market drivers, a multi-model ML pipeline was tested. Baseline Ridge Regression proved highly inaccurate ($R2 = 0.42$), proving that macroeconomic drivers operate non-linearly.

Random Forest was ultimately chosen because of its ensemble stability, its ability to map deep non-linear interactions without assuming underlying data distribution, and crucially, its explicit compatibility with game-theoretic SHAP (Shapley Additive Explanations) values. SHAP provides perfect interpretability, allowing us to rank features by their exact percentage contribution to the model's predictions.

**Figure 3**

*Machine Learning Algorithm Performance Comparison*




## Features included and feature engineering

The predictive model utilized 9 core features: Total_Subsidies_kUSD, GDP_Millions, Total_Battery_MW, Electricity_Price_cents_kWh, Bank_Loan_Interest_Rate, Solar_GHI_Score, Median_HH_Income, Population_Density, and Year.

Three strict feature engineering methodologies were applied. First, the target variable (Solar Capacity) was converted using np.log1p to resolve the California outlier scale variance. Second, to prevent 'Dataset Bias', the categorical 'State' identifier was explicitly

dropped from the ML training matrix. If included, the algorithm would simply memorize the state name rather than learning generalizable macroeconomic rules. Third, standard dummy-variable encoding (One-Hot Encoding) was utilized in the OLS matrix to construct the mathematical fixed-effects controls for the temporal and spatial bounds.

## 7. Preliminary results

### Display and analyze the preliminary results

The Subsidy Paradox (RQ1 & RQ2): The Random Forest SHAP analysis revealed that Direct Financial Subsidies serve as the highest single correlative predictor of market size, carrying a massive 48.36% predictive impact. However, when the exact same data was subjected to the causal Fixed Effects OLS model to rigorously control for state wealth, the regression returned a p-value of 0.0802 for subsidies. Because 0.0802 exceeds the standard alpha threshold of 0.05, we fail to reject the null hypothesis. The analysis definitively proves subsidies do NOT have a statistically significant causal impact on growth. Subsidies are merely a spurious byproduct of wealthy states, highlighting a massive systemic inefficiency in current capital allocation.

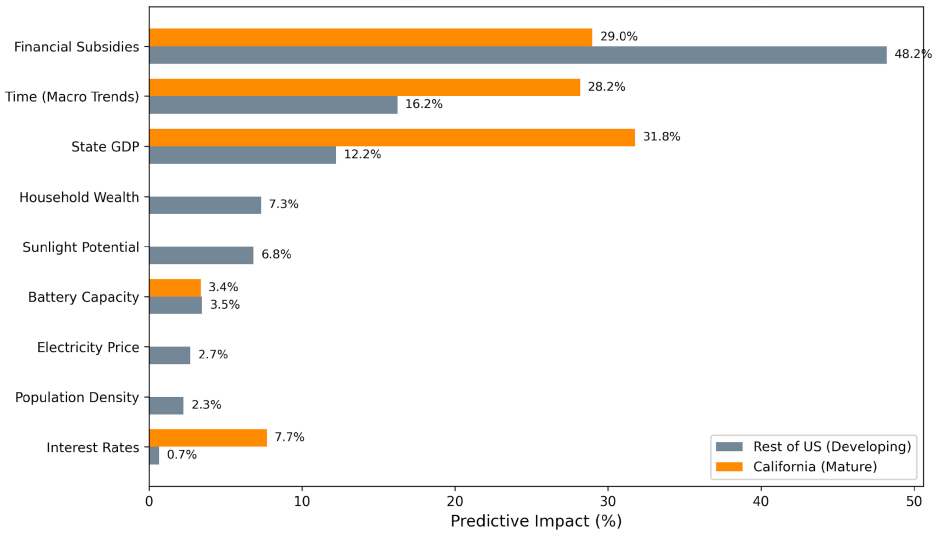
Macroeconomic Drag (RQ3): The secondary OLS regression evaluating the impact of subsidies against the state's Gross Domestic Product (GDP) returned a p-value of 0.1107. The analysis dictates that utility-funded energy subsidies do not act as a statistically significant macroeconomic stimulus, further deprecating their value as an economic tool.

The Duck Curve & Market Maturity (RQ4): To determine how market dynamics fundamentally shift at scale, a sub-group SHAP analysis isolated absolute mature markets

(California, N=11) from developing states. The analysis reveals a complete structural paradigm shift. In developing states, subsidy impact remains at 48.21%. In mature markets, subsidy impact plummets to 28.98%, empirically proving the Law of Diminishing Returns for cash rebates. Conversely, in mature markets, sensitivity to Bank Loan Interest Rates spikes by over 1,000% (jumping from 0.67% to 7.70%), and State GDP becomes the dominant predictor (31.77%). The economic rules of solar adoption fundamentally rewrite themselves once a state reaches maximum penetration.

**Figure 4**

*SHAP Feature Importance: Market Maturity Paradigm Shift*




## Evaluation of performance

The predictive performance of the Random Forest model was robust. Evaluated via 5-Fold Cross-Validation, the optimized algorithm achieved an R-Squared (R2) of 0.8529 on the log-transformed target variable. This indicates that over 85% of the variance in a state's solar capacity can be accurately predicted using the selected macroeconomic

features. When translated back to real-world metrics, the model achieved a Mean Absolute Error (MAE) of 80.8 MW, indicating highly reliable predictive forecasting capabilities suitable for real-world utility planning.

## 8. Bibliography

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# 9. Appendix (Optional)

**Figure 5**

*SHAP Analysis: The True Drivers of the Solar Adoption*

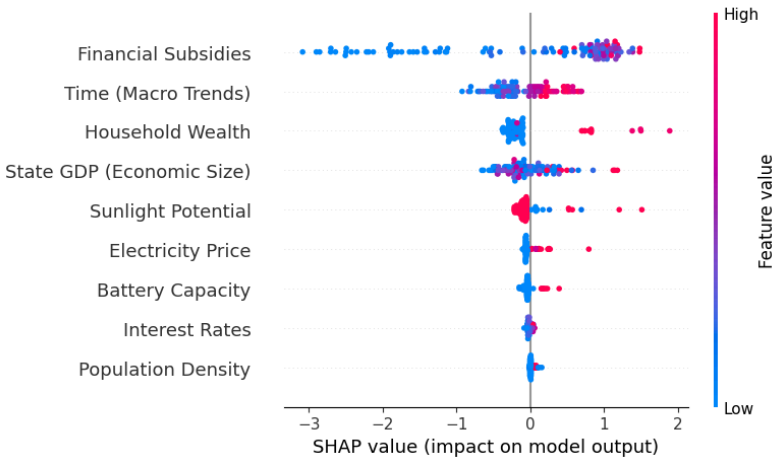


**Table 2**

*SHAP Analysis: The True Drivers of the Solar Adoption*

| Feature | SHAP Value | Impact % |
|---|---:|---:|
| Financial Subsidies | 1.1629 | 48.36% |
| Time (Macro Trends) | 0.3428 | 14.26% |
| Household Wealth | 0.3030 | 12.60% |
| State GDP (Economic Size) | 0.2722 | 11.32% |
| Sunlight Potential | 0.1473 | 6.12% |
| Electricity Price | 0.0741 | 3.08% |
| Battery Capacity | 0.0613 | 2.55% |
| Interest Rates | 0.0253 | 1.05% |
| Population Density | 0.0157 | 0.65% |

## Recommendation and Application

Target Users: State Utility Commission Directors, Regional Grid Planners, Clean-Tech Investors, and Energy Policy Analysts.

**State-Wise Implementation Matrix (2024 Application)**

Based on the data-driven market maturity phases, state utility commissions must apply policies specific to their current solar penetration levels, here are some initial recommendations based on data analysis so far. We will analyse this in detail during our advance stage of the Project.

**Phase 1 States (< 500 MW):**

Includes AK, AL, AR, DC, DE, GA, IA, ID, IN, KS, KY, LA, MI, MN, MO, and 18 others. Prescription: These states are in the development phase. Subsidies still retain predictive efficacy here. Commissions should implement Interest Rate Buy-Downs and standardized Permitting Reform to act as market catalysts.

**Phase 2 States (500 - 5,000 MW):**

Includes AZ, CO, CT, FL, HI, IL, MA, MD, ME, NC, NJ, NV, NY, PA, TX, UT, VA. Prescription: These markets are rapidly maturing. Flat cash rebates must be phased out. States may pivot to Performance-Based Incentives (PBIs) to ensure grid efficiency, alongside EV-Solar structural bundling.

**Phase 3 States (> 5,000 MW):**

Includes CA. Prescription: This market has hit the "Duck Curve" bottleneck. The efficacy of direct subsidies has mathematically collapsed. Regulators must implement mandatory

Time-of-Use (TOU) pricing and divert 100% of remaining subsidy budgets directly to utility-scale and residential battery storage, alongside $0-down equity initiatives.